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Research and design of a high-security configurable RO-PUF based on FPGA

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Abstract

Aiming at the lack of reliability and uniqueness of the traditional configurable ring oscillator physical unclonable function (RO-PUF) due to the simple configurable path, a configurable RO-PUF is proposed based on the delay characteristics of the ring oscillator. The design scheme redesigned the RO structure, making the configurable path of the RO-PUF more flexible and increasing the number of incentive response pairs (CRPs); At the same time, the introduction of a lightweight SPONGENT hash algorithm in the RO-PUF structure makes the mapping between incentives and responses more complicated, prevents attackers from building model attacks by acquiring CRPs, and enhances RO-PUF security. And then, the design scheme is implemented on Xilinx's Spartan series FPGA chip, and the experimental results are compared and analyzed. The results show that the configurable RO-PUF designed with this scheme has an average reliability of 98.87% and uniqueness of 49.13%.

Keywords: RO-PUF; Ring oscillator; SPONGENT; FPGA; CRPs

1. Introduction

With the rapid development of integrated circuit technology and the popularization of electronic equipment, various information security issues such as malicious insertion of Trojan horse files, counterfeit device labels, and counterfeit chips in integrated circuits have arisen. In order to protect user data and product information, Electronic products' anti-counterfeiting encryption technology is particularly important [1-2]. Physical Unclonable Function (PUF) provides new ideas for solving such information security problems [3]. RO-PUF is mainly composed of

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several odd-numbered inverters connected [4-5]. The oscillation signal of the entire link is relatively stable, and the circuit structure has no symmetry requirements, which is easy to implement in FPGA. Therefore, this article selects RO-PUF as the research and improvement object.

Firstly, based on the delay characteristics of RO-PUF, this paper proposes an improved configurable RO-PUF method based on RO grouping. By increasing the configurable structure of RO-PUF, the reliability and uniqueness of RO-PUF is improved. Secondly, the lightweight SPONGENT hash function is introduced to process the RO-PUF's input stimulus, removing the correlation between the output and the physical details of the RO-PUF measurement, making the stimulus random and increasing the difficulty of modeling attacks. Finally, the response generated by the RO-PUF is subjected to error correction processing to minimize the bit error rate.

2. Principle Introduction

2.1. RO-PUF principle

RO-PUF is mainly composed of ring oscillator, multiplexer, counter, etc[6]. Its basic structure is shown in Fig.1. The data selector MUX receives the output signals of the two ring oscillators RO in the N oscillation circuits and the external input excitation signal of the system. The two counters respectively count the oscillation frequencies of the two ROs in the same time period, and finally output through the comparator. Oscillation frequency result, the result is logic 1 or 0.

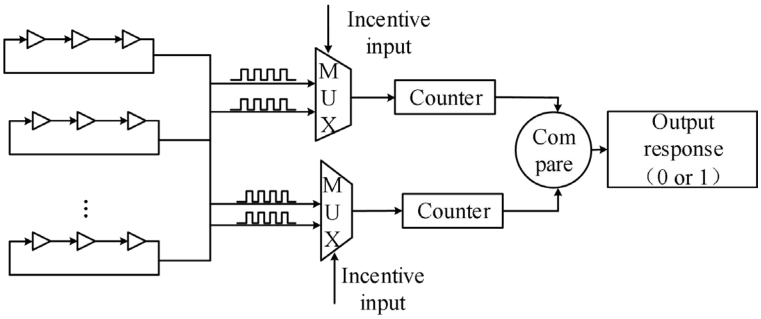


Fig. 1. RO-PUF basic structure

The structure of the traditional configurable RO-PUF is shown in Fig.2. This configuration method obtains a specific oscillation frequency by changing the input excitation signal, so that the output frequency signal has a higher safety. However, the configurable RO-PUF RO structure implemented by the traditional method is simple, the number of CRPs generated is not enough, the reliability and uniqueness of the circuit designed with it are not high, and the input excitation signal is not randomized, and the generated excitation is not with randomness, an attacker can build an attack model by acquiring CRPs. Based on this, this paper proposes a configurable RO-PUF design method based on RO grouping, which improves the configurability of RO-PUF, and introduces a lightweight SPONGENT hash algorithm to prevent attackers from modeling attacks on CRPs.

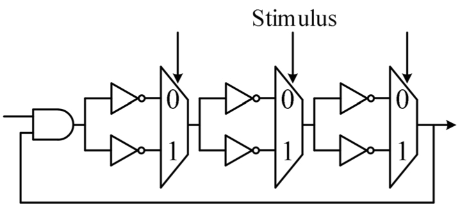


Fig. 2. Traditional configurable RO structure

2.2. Principle of Lightweight SPONGNT Hash Algorithm

Lightweight SPONGENT is a sealing sponge strategy based on a wide range of PRESENT-type arrangements, which depends on the sponge structure [7-8] as shown in Fig.3.

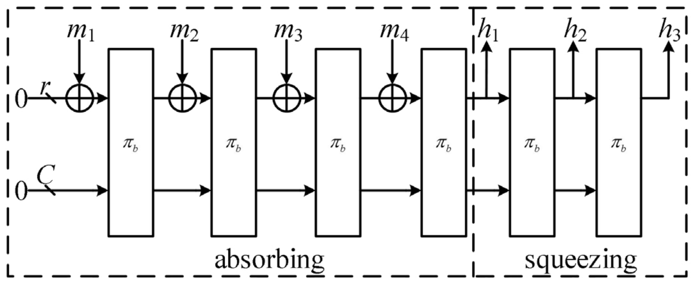


Fig. 3. SPONGENT sponge structure

Where *mi* is the input, h_i is the hashed output, *r* is the rate, *c* is the capacity, *πb* is the permutation function, and the size of the internal state *b* is the permutation width, satisfy $b = r + c \geq n$, *n* is the output size. In lightweight SPONGENT, bit 0 is used as the initial value before absorbing. The message is first filled with a single 1 bit, followed by a certain number of 0 bits, until it is a multiple of *r*, and then it is cut into *r* bit message blocks. It is stored in the first *r* bits of state (state can be regarded as a three-dimensional array). Once all message blocks are absorbed, the first *r* bits of state are used as output and are replaced by *πb* until *n* bits are returned. The use of a lightweight SPONGENT hash algorithm in the RO-PUF makes the excitation of the input RO-PUF random, ensuring the stability of the RO-PUF.

3. Design Section

The configurable RO-PUF based on the lightweight SPONGENT hash algorithm actually physically binds the configurable RO-PUF and the encryption algorithm in an inseparable manner [9], preventing attackers from building model attacks by acquiring CRPs. The error correction algorithm is used to reduce the bit error rate of the RO-PUF response. The RO-PUF response after the error correction algorithm is processed by the lightweight SPONGENT hash function, which releases the physical association between the output and the RO-PUF. Modular attacks are more difficult. Add controllable personality parameters to Control-Logic and participate in the SPONGENT hash calculation of the input to prevent attackers from maliciously tracking the user's private information using unique identifiers. The top design of RO-PUF is shown in Fig.4.

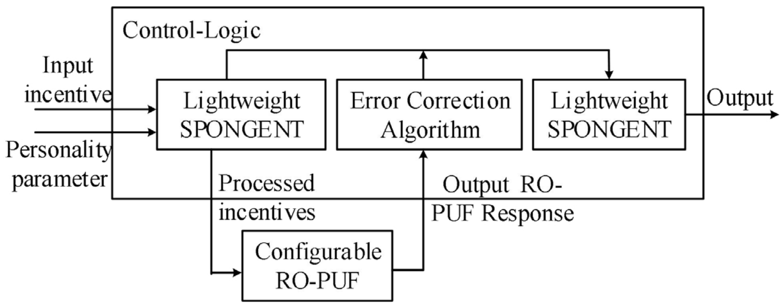


Fig. 4. Configurable RO-PUF top-level diagram based on lightweight SPONGENT hash algorithm

3.1. Optimized design of configurable RO-PUF

In order to increase the complexity of the RO structure and generate a larger number of CRPs, this paper proposes a RO-PUF configurable structure based on RO grouping. In this structure, the ROs in the configurable RO-PUF are first divided into several groups. If there are N-ROs in the RO-PUF, the ROs are divided into N/2 groups. Configure performance. The overall structure of the configurable RO-PUF requires that the macro structure of the RO be consistent, that is, the effects of process deviations are excluded. The number and structure of the ROs must be exactly the same. Based on this principle, this article uses the 7-stage RO as an example to design a group of configurable RO- PUF, its packet structure is shown in Fig.5.

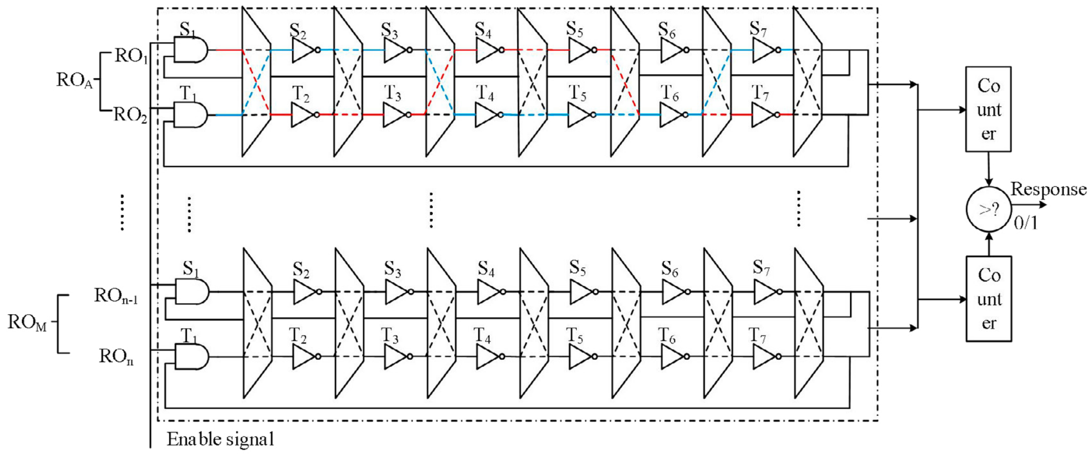


Fig. 5. 7-stage RO structure design

In the 7-stage RO structure, the physical components in an RO group include AND gate, multiplexer, and inverter. and the multiplexer is used to plan the RO path. The basic idea of grouping based on RO is to divide RO_1 and RO_2 into one group, RO_3 and RO_4 into one group, and so on. N-ROs are divided into N/2 groups. Each group after the grouping is regarded as a new RO, and then structure the interior of the grouped RO.

Assume that when the enable signal is 1, the counter receives the output frequency of RO_1, and when the enable signal is 0, the counter receives the output frequency of RO_2. Each multiplexer chooses to receive the output frequency of RO_1 or RO_2 according to the input signal. AND gates and six inverters have a total of 2^7 paths. For example, when S_1-S_7=1001100, RO path is $S_1 \rightarrow T_2 \rightarrow T_3 \rightarrow S_4 \rightarrow S_5 \rightarrow T_6 \rightarrow T_7$, the data flow direction is the red line in the figure, when S_1-S_7=0110001, RO path is $T_1 \rightarrow S_2 \rightarrow S_3 \rightarrow T_4 \rightarrow T_5 \rightarrow T_6 \rightarrow S_7$, data flow is blue in the figure. A newly configured RO in this structure has a total of 128 paths, which is 9 times that of traditional RO-PUF.

3.2. Processing of RO-PUF incentive and response

The number of RO-PUF configurable paths based on RO groups increases exponentially, which makes the number of RO-PUF CRPs significantly increase. In order to prevent attackers from establishing attack models by acquiring CRPs and reducing the bit error rate of RO-PUF responses, this design Introduce lightweight SPONGENT hash algorithm and response error correction algorithm [10-11].

Suppose a replacement-based sponge structure has $n \geq c$, $c/2 > r$, and satisfies the parameter selection of all sponge variables, the complexity of the modeling attack is $2n - r + 2c/2$. The RO-PUF input incentives that do not use a hash function are difficult to predict. The possibility of being subjected to modeling attacks is as high as 30%. The lightweight SPONGENT hash algorithm makes the input RO-PUF incentives random, it is difficult for an attacker to establish a mathematical model to attack RO-PUF, and the attack rate is only about 2%.

In this paper, concatenated codes are used to correct the RO-PUF output response. The error correction processing diagram is shown in Fig.6. First use the error correction code Repetition Code as the internal encoding, that is C (REP), to reduce the error rate of the codeword to a relatively low level, and then use a code C (BCH) with a stronger error correction capability than C (REP) as the external encoding to perform the final error correction. It is assumed that the probability of an error occurring in each bit of the RO-PUF response is p, and the error rate of the response processed by the error correction algorithm does not exceed 5%, and the response without the error correction processing of the EC algorithm, p will reach more than 10%.

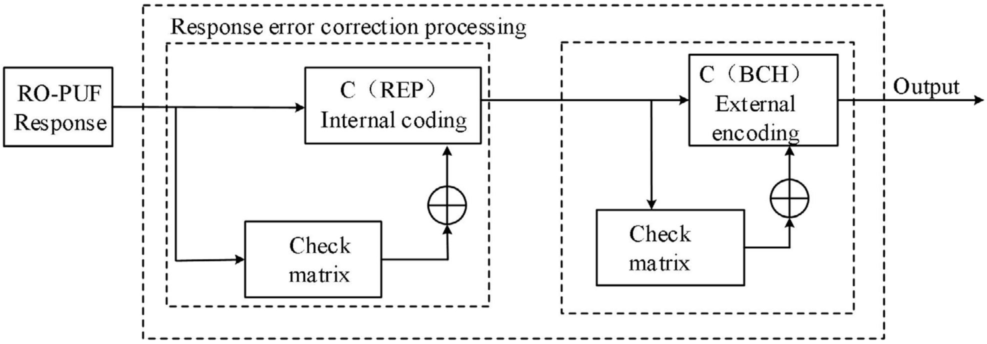


Fig. 6. Response error correction processing diagram

4. Testing and analysis

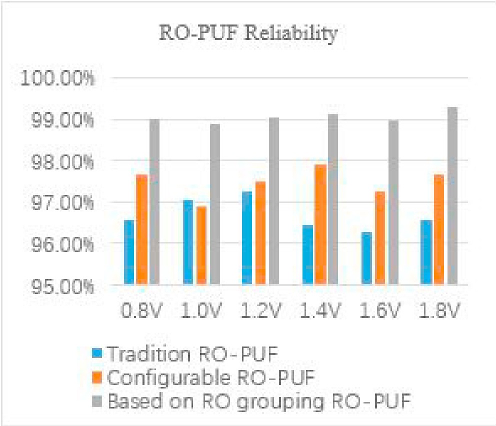
4.1. Testing platform

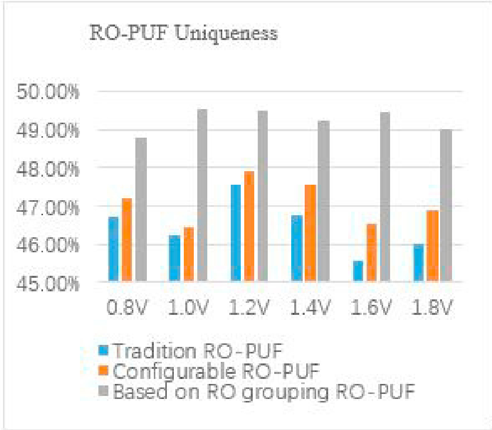
The RO-PUF design based on RO grouping is finally implemented on Xilinx's Spartan series FPGA chips. The test language uses Verilog HDL. In order to maintain the consistency of the structure between each level of RO, the RO is packaged into hard macros and placed on the top layer. The test circuit is instantiated in the module. After the FPGA board is powered on, the test is performed by the keys on the board. The logic analyzer collects the specified pin signals and displays them on the host computer excel files and statistics by MATLAB [12].

4.2. Reliability and uniqueness testing

Reliability and uniqueness are important performance indicators of RO-PUF. RO-PUF reliability test is to test the same RO-PUF under the same environment with the same stimulus, collect the response in the changed environment and compare the response value in the ideal environment, and calculate the on-chip Hamming distance. To determine the RO-PUF reliability, the closer the Hamming distance value of the RO-PUF chip is to 0%, it indicates that the designed RO-PUF reliability is higher. The uniqueness test of RO-PUF is generally to input stimulus to multiple chips with the same RO-PUF structure, and calculate the inter-chip Hamming distance between responses generated by the same input stimulus. The closer the distance value is to 50%, the more unique high.

The environmental factors that affect the reliability and uniqueness of RO-PUF are temperature, operating voltage, and magnetic field. Due to limited test conditions, this experiment tested the reliability and uniqueness of RO-PUF based on RO grouping at different voltages. The voltage value is set to 0.8V, 1.0V, 1.2V, 1.4V, 1.6V, 1.8V. At the same time, compared with traditional configurable RO-PUF and traditional RO-PUF, the results are shown in Fig.7. Under different voltages, the reliability of RO-PUF based on RO grouping is stable. It can be seen from the histogram (a) in Fig. 7 that the average on-chip Hamming distance of the RO-PUF designed by this method is 1.13%, the reliability is 98.87%, which is almost close to 100%. The method is more reliable; meanwhile, as shown in Fig.7(b), the average Hamming distance between RO-PUF-based RO-PUF slices is 49.13%, which is close to 50%, which is 47.61% of configurable RO-PUF and traditional RO. Compared with -PUF 46.82%, RO-PUF based on RO grouping has stronger uniqueness.





(a) Effect of voltage on RO-PUF reliability (b) Effect of voltage on RO-PUF uniqueness

Fig. 7. Results comparison chart

5. Conclusion

This paper proposes an improved configurable RO-PUF structure based on RO grouping, making the configurable path of RO-PUF more flexible, improving the reliability and uniqueness of RO-PUF, and applying SPONGENT lightweight hash function in the RO-PUF structure, the RO-PUF has a larger CRPs space, improves the randomness of the input excitation, and makes modeling attacks more difficult. In the end, the reliability and uniqueness of RO-PUF were simulated under different voltages and temperatures. The experimental results show that RO-PUF based on RO grouping has better performance than traditional designs and has certain practical significance.

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